hdoc day 48 by trishil chauhan

Problem Statement

Read a variable-length integer array and pass its address to **pivot_index()**. Find the leftmost index where the sum of elements strictly to its left equals the sum strictly to its right. Print -1 if no pivot exists.

C Program

```
#include <stdio.h>

void pivot_index(const int *arr, int n) {
    long long total = 0, left = 0, right;
    int i;

    for (i = 0; i < n; i++)
        total += *(arr + i);

    for (i = 0; i < n; i++) {
        right = total - left - *(arr + i);
        if (left == right) {
            printf("Pivot index: %d\n", i);
            printf("Left sum: %lld\n", left);
            printf("Right sum: %lld\n", right);
            return;
        }
        left += *(arr + i);
    }
    printf("Pivot index: -1\n");
}

int main(void) {
    int n, i;
    printf("Enter array size: ");
    if (scanf("%d", &n) != 1 || n <= 0) {
        printf("Invalid array size.\n");
        return 1;
    }

    int arr[n];
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            printf("Invalid element.\n");
            return 1;
        }
    }
    pivot_index(arr, n);
    return 0;
}
```

Sample Output 1 - Pivot exists

```
Enter array size: 6
Enter 6 elements: 1 7 3 6 5 6
Pivot index: 3
Left sum: 11
```

```
Right sum: 11
```

Sample Output 2 - No pivot

```
Enter array size: 3
Enter 3 elements: 1 2 3
Pivot index: -1
```

Explanation

First, calculate the total sum. For each index i, compute right = total - left - arr[i]. If left equals right, print that index and return immediately, ensuring the leftmost match. Otherwise add arr[i] to left and continue. Initially left is zero; for the final element, right is zero.

Complexity: O(n) time and O(1) auxiliary space (excluding the input variable-length array).